

Integrating MATSim and Microscopic Railway Simulation for Passenger Transport Digital Twinning

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Introduction

Integrating and prioritising public and active transport modes is critical to reduce transport emissions in global cities. Multimodal services need to be tuned to societal demands and offer easy, reliable, and affordable access to encourage modal shift from private vehicles (Steele et al., 2026). However, multimodal transport simulation represents a highly complex adaptive system, requiring integration of features such as representative community agent populations, system dynamic analysis, intrinsic network priorities and event-based simulation. This has led to the creation of tools capable of representing both large-scale travel behaviour and detailed operational dynamics.

Agent-based modelling has become an important approach in transport research, supporting the analysis of travel demand, infrastructure changes, and multimodal mobility systems (Bastarionto et al., 2023). One of the most widely used tools is MATSim, an open-source agent-based transport simulation framework that models travellers as autonomous agents optimising route, departure time, and mode choice (Horni et al., 2016; Bastarionto et al., 2023). MATSim is widely used for large-scale behavioural and multimodal simulation. However, in most cases, its representation of rail operations remains highly abstract.

Railway systems exhibit complex interactions between infrastructure, signalling, rolling stock, and service patterns, resulting in emergent operational behaviours such as congestion, delay propagation, and capacity constraints. These effects are difficult to capture using timetable-based assignment approaches, where train movements are typically represented at an abstract level within MATSim. This abstraction limits the ability to represent how railway operational constraints influence service reliability and passenger decision-making. While recent developments have introduced greater operational detail (Kaddoura et al., 2024), microscopic railway dynamics remain outside the scope of standard MATSim representations. This motivates the integration of a dedicated microscopic railway simulator, enabling a more realistic representation of rail operations within a federated digital twin framework for urban mobility (Li et al., 2026a).

This paper presents a hybrid MATSim/BRaVE framework developed within the TransiT Challenge-Led Demonstrator 2 (CLD2), focused on passenger transport digital twinning for the West Midlands. The framework has been demonstrated within CLD2 through a pilot application on the Cross City Line corridor, connecting the University of Birmingham and Queen Elizabeth Hospital within central Birmingham.

MATSim/BRaVE Integration Context

BRaVE (Birmingham Railway Virtual Environment) is a microscopic railway simulation platform that models train movements using detailed infrastructure, signalling, interlocking, train physics, and timetable data (Douglas et al., 2016). Unlike timetable-based approaches, BRaVE explicitly represents operational interactions such as route conflicts, signal control, capacity constraints, and delay propagation, providing a high-fidelity representation of railway operations.

MATSim Capability	BRaVE Capability
Agent travel demand	Train movement dynamics
Route and mode choice	Signalling and interlocking
Departure time choice	Dispatching and conflict resolution
Mutimodal trip construction	Infrastructure and capacity constraints.
Passenger behaviour	Delay propagation and reliability.

Table 1: MATSim and BRaVE Capabilities

The coupled framework combines MATSim and BRaVE into a synchronized simulation environment. MATSim is extended via a custom *MobsimEngine* plugin, including a departure handler that intercepts rail *Leg PlanElements*. When a rail leg begins, the agent is transferred to BRaVE, where a custom dispatcher handles platform assignment and allocation to feasible trains. Upon arrival at the destination, the agent is returned to MATSim, where the leg is marked as completed.

Synchronization is achieved through MATSim's *MobsimEngine doSimStep* method, which acts as the master simulation clock. Each call advances BRaVE's simulation clock, with *doSimStep* blocking until BRaVE completes its corresponding simulation step. This ensures both systems progress in lockstep and maintains temporal consistency between the two systems.

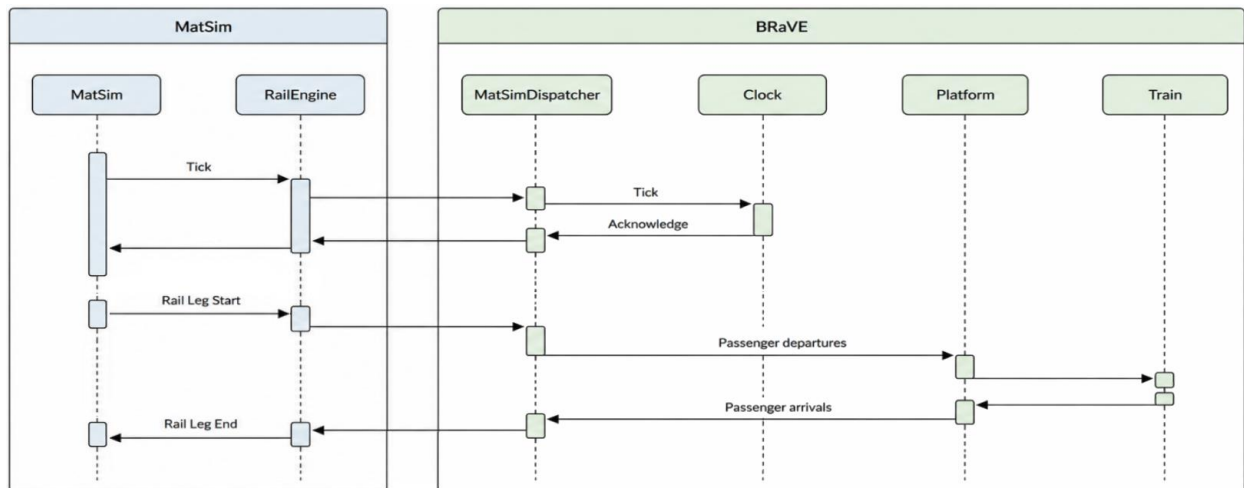


Figure 1: MATSim BRaVE Sequence diagram

Outlook and Conclusions

The MATSim/BRaVE integration provides a hybrid simulation framework combining large-scale travel demand with microscopic rail operations. Their synchronous execution enables interaction between passenger demand and operational constraints within a unified modelling environment. Within the TransiT CLD2 programme, the framework has been implemented to assess how improved utilisation of rail infrastructure and services can support mode shift from higher-carbon transport

modes, thereby contributing to transport decarbonisation in the West Midlands. Key applications include:

- **Capacity-constrained rail operations:** evaluating the extent to which existing infrastructure can accommodate increased rail demand arising from mode shift away from higher-carbon transport modes, while identifying operational bottlenecks that may constrain utilisation.
- **Rail investment and timetable strategies for net-zero pathways:** combining behavioural responses with operational constraints to test infrastructure and service interventions in terms of mode shift and emissions reduction potential.
- **Reliability effects on travel behaviour:** using BRaVE to represent operational disruptions and MATSim to capture resulting changes in passenger choice and cost.

More broadly, the approach demonstrates a method for federating complementary transport simulators within the TransiT CLD2 digital twin architecture. As the multi-level simulation component of the federation, the integrated MATSim/BRaVE framework provides behavioural and operational evidence that can iteratively inform the Minimum Viable Ontology (MVO), enriching the shared semantic representation of network conditions, passenger behaviour, and rail operations (Li, et al., 2026b).

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References

- Bastarionto, F.F., Hancock, T.O., Choudhury, C.F., Manley, E. (2023). Agent-based models in urban transportation: review, challenges, and opportunities. *European Transport Research Review*, 15, 19. <https://doi.org/10.1186/s12544-023-00590-5>
- Douglas, H., Weston, P., Kirkwood, D., Hilmansen, S., Roberts, C. (2016). Method for validating the train motion equations used for passenger rail vehicle simulation. *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*, February 15. <https://doi.org/10.1177/09544097166317>
- Horni, A., Nagel, K., Axhausen, K. (2016). *The Multi-Agent Transport Simulation MATSim*. Ubiquity Press, London.
- Kaddoura, I., Unterfinger, M., Hettinger, T., Rakow, C., Rieser, M. (2024). A large-scale hybrid micro- and mesoscopic simulation approach for railway operation. *Procedia Computer Science*, 238, 714–721. <https://doi.org/10.1016/j.procs.2024.06.082>
- Li, J., Grübel, J., Nadi, A., Snelder, M., van Arem, B., Gao, J. (2026a). Digital twin federation for urban mobility assessment: Definition, pillars, and a human-in-the-loop functional architecture. *Transportation Research Part A: Policy and Practice*, 211, 105086. <https://doi.org/10.1016/j.tra.2026.105086>
- Li, J., Steele, H., Preece, J., Kirkwood, D., Sun, S., Jiang, W., Logan, A., Das, S., Almeida, R., Sevegnani, M., Easton, J., Roberts, C., Flynn, D., Greening, P. (2026b). A Challenge-Led Digital Twin

Federation Architecture for Urban Passenger Transport Decarbonisation. *Procedia Computer Science*, 280, 303–311. <https://doi.org/10.1016/j.procs.2026.04.040>

Steele, H., Duvnjak, J., Byron, P., Matyas, M., Easton, J., Roberts, C., Flynn, D., Greening, P. (2026). A co-created framework to define digital twinning use cases for urban transport decarbonisation. *Urban Science*, 10(3), 140. <https://doi.org/10.3390/urbansci10030140>